P03

Ant Colony Optimization Applied to Dynamic Network Routing

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Domain DOM_NET

Abstract

We introduce AntNet, an adaptive distributed routing algorithm based on the ant colony optimization meta-heuristic. Mobile routing agents explore communication subnets and adapt routing tables using backward reinforcement, yielding high throughput and robust congestion avoidance.

Abstract quoted from G. Di Caro et al., Journal of Artificial Intelligence Research (1998).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?